Blurring/Clutter Mitigation in Quarry Monitoring by Ground-Based Synthetic Aperture Radar

Lapo Miccinesi, *Member, IEEE*, Alberto Michelini, and Massimiliano Pieraccini, *Member, IEEE*

***Abstract*— Ground-based synthetic aperture radar (GBSAR) systems are widely used for monitoring slopes, especially in quarries and open pits. Unfortunately, the movement of the machinery (cranes, trucks, shovels, and excavators) in the radar field of view often prevents to correctly focus large circular sections of the radar images, producing a characteristic artifact, often named "blurring." Therefore, the aim of this article is to propose both an acquisition modality and an image processing technique able to mitigate the blurring due to moving clutter. The proposed acquisition modality (named "free-running" or "on-the-fly") allows to filter out the high-frequency clutter, which is often noticeable in GBSAR images, but it is not effective with low-frequency clutter caused by slow movement of machinery like heavy trucks or large cranes. For this reason, the authors propose even a suitable processing method that can be used in combination with high-frequency sampling. The application of these two methods has been proved to be very effective in realistic quarry scenario.**

***Index Terms*— Clutter, Doppler, ground-based synthetic aperture radar (GBSAR), quarry, radar, synthetic aperture radar.**

I. Introduction

GROUND-BASED synthetic aperture radar (GBSAR) systems [1] are popular remote sensing instruments for short-term [2] and long-term [3] monitoring of landslides, glaciers [4], and mines [5].

A GBSAR is able to synthesize a radar image by exploiting the movement of a radar front-end along a linear rail [6] or along an arc [7], assuming that no appreciable movement occurs in the monitored scenario during the acquisition time of each single image. Indeed, any displacement that has a range component larger than a consistent fraction of the wavelength (e.g., one quarter of wavelength) causes a typical clutter that affects all the neighboring pixels lying in the same radial range. In a radar image in the Cartesian coordinates, this clutter appears like a blurred circular crown. This effect has been noticed since the earliest experiments of GB-SAR systems [8]. Recently (2018), Lort *et al*. [9] experimentally studied the cross-range blurring effect and the reduction of coherence caused by wind. They suggest correctly to reduce the scanning time while maintaining the total transmitted energy for mitigating these effects.

Currently, many hundreds of GBSAR systems are installed worldwide as safety equipment for monitoring the slope stability of large quarries [10]. Monitoring the slope close to the operating machinery is a priority for the safety of workers. Unfortunately, the movement of the machinery (cranes, trucks, excavators, etc.) in the radar field of view can blur large circular sections of the radar image.

Therefore, the aim of this article is to propose an acquisition modality and a processing strategy able to mitigate the blurring due to moving clutter.

Standard GBSAR systems move along a linear rail (or along an arc) step by step (named also "stop and go"), so the radar front-end is motionless during each single acquisition. An alternative operating modality is to acquire continuously during the movement ("free-running" or "on-the-fly"), keeping constant the speed, similar to a spaceborne SAR moving along its orbit. The unique advantage of this modality is that with a suitable sampling frequency, the clutter due to moving targets is rejected by the focusing algorithm itself which effectively operates as a matched filter [11]. This modality does not require to reduce the scanning time, so it allows to keep constant the transmitted energy without increasing the microwave power. A further advantage of not increasing the speed or the radar along its guide is less stress for the mechanical components, which in this equipment are always subject to heavy wear.

High sampling frequency allows to cut off high-frequency clutter that could refold inside the acquisition band. Unfortunately, it cannot do anything for clutter due to scatters moving slowly producing frequency components in the band of the Doppler filter. This is not a rare occurrence in the quarry scenario, where large and slow machineries are moving. As the blurring is due to the movement of the clutter during the scan of the radar head along the rail, a simple way to reduce it is to apply a heavy windowing in the cross-range (i.e., along the scan) that "virtually" reduces the scan length. The evident disadvantage of this approach is the worsening of resolution.

For this reason, we propose a special processing method that can be used with GBSAR operating both in step-by-step modality and in free-running modality, which mitigates the blurring due to clutter slowly moving without worsening the cross-range resolution. As it will be better described below, the basic idea of this processing technique is to identify and cut off the components of the signal (before the cross-range

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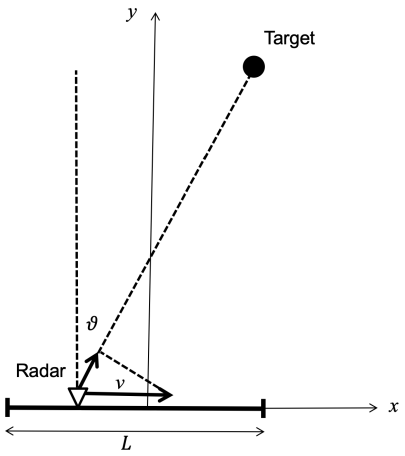


Fig. 1. Working principle of a free-running linear GBSAR system. The radar front-end moves along the x-axis at speed v.

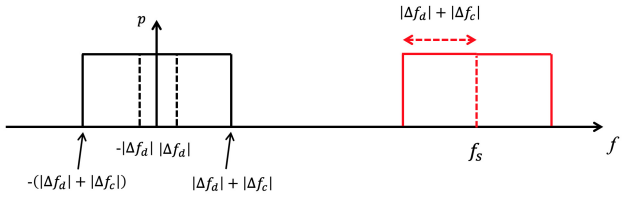


Fig. 2. Sampled Doppler spectrum of the signal acquired by a free-running GBSAR.

focusing) that produce the largest differences between acquisitions at different times. The application of these two methods has proved to be very effective in realistic quarry scenarios.

It is worth to spend a word about the atmospheric correction that is a key step in every interferometric processing, to detect correctly the real displacement. Atmospheric changes occur in times much longer that a single image or a couple of images. Therefore, we can reasonably affirm that blurring/clutter mitigation procedures reported in this article can be applied independently of the atmospheric correction.

II. High Sampling Frequency Acquisition Method

Fig. 1 shows a sketch of the working principle of a free-running linear GBSAR. The radar front-end moves along the x-axis at speed v.

The azimuth ϑ of a stationary target in the field of view is retrieved by its Doppler shift Δf_d

$$\Delta f_d = 2\left(\frac{v}{c}\right) f_c \sin(\vartheta) \tag{1}$$

where f_c is the central frequency of the transmitted signal, and c is the speed of light. Any moving target (clutter) will have an additional frequency shift Δf_c, and therefore, the Doppler spectrum of the signal acquired by the radar is $\pm(|\Delta f_d| + |\Delta f_c|)$. To simplify the following discussion, the spectrum of the clutter is supposed flat with spectral density p, as shown in Fig. 2. When the signal is sampled at frequency f_s, the spectrum has replica as shown in Fig. 2.

The Doppler image processing acts as a matched filter between $\pm|\Delta f_d|$. Therefore, if $f_s > (2|\Delta f_d| + |\Delta f_c|)$, the replica does not fold in the Doppler filter $(\pm|\Delta f_d|)$, and

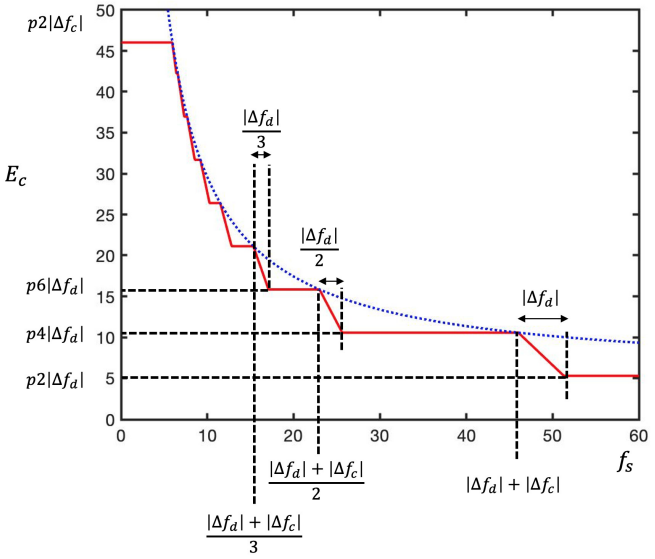


Fig. 3. Clutter energy inside the Doppler filter versus sampling frequency (for a flat clutter spectrum).

the clutter signal energy (E_c) is $2p|\Delta f_d|$. When $f_s < (2|\Delta f_d| + |\Delta f_c|)$, the edge of the replica overlaps the Doppler filter, and therefore, the clutter energy increases linearly until $f_s = (|\Delta f_d| + |\Delta f_c|)$. For this value of sampling frequency, the spectrum folds twice, but the edge of the second folding does not overlap the Doppler filter until $f_s = (2|\Delta f_d| + |\Delta f_c|)/2$ and so on, as shown in Fig. 2.

The dotted line plotted in Fig. 3 is

$$E_c(f_s) = \frac{2p|\Delta f_d|(|\Delta f_d| + |\Delta f_c|)}{f_s} + 2p|\Delta f_d|. \tag{2}$$

So, the clutter energy can be stated roughly inversely proportional to the sampling frequency. This is a result rather general that should not depend on specific clutter spectrum. Indeed, in the experimental section, Pieraccini *et al.* [12] demonstrate that the inverse of (peak signal–noise ratio (PSNR) is approximately proportional to the inverse of sampling frequency.

III. Coherent Change Subtraction Method

As stated above, high sampling frequency is able to cut off high-frequency clutter that could re-fold inside the Doppler band. Unfortunately, high-frequency clutter (e.g., due to engine vibration and moving shovels) is not the only kind of clutter that can be observed in a typical quarry scenario. Slow machinery like large rotating cranes or bulldozers that move from place to place produce a troublesome low-frequency clutter. This second kind of clutter cannot be mitigated by high-frequency acquisition.

Therefore, the approach the authors propose for rejecting/mitigating this kind of clutter is the following: 1) two radar images are acquired at different times (typically a few minutes) and they are processed only in range (in the case of an FM radar, it means to calculate the fast Fourier transform (FFT) of each single sweep); 2) the image difference between the two semi-processed images is calculated; 3) an amplitude threshold

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Fig. 4. Radar equipment installed in front of the Pedogna quarry, Italy.

TABLE I
RADAR PARAMETERS

Central frequency	17.20 GHz
Band	400 MHz
Field of view (-3dB)	50 °
Length of the rail	1.8 m
Sampling frequency	625 Hz
Radar-head speed along the rail	0.03 m/s

is defined, and all the pixels whose change between the two images overcomes the threshold is made equal to zero; and 4) finally the image is processed along the cross-range.

Note that as the image difference is calculated between matrices of complex numbers, this subtraction takes into account the coherence changes between the two images.

The key idea of the process in four steps above described is that pixel removal operates in the range domain before integration along the x-axis. A single pixel in the x–y image is the integration of several acquisitions at different points along the mechanical rail at different times (as the radar head scans along the rail in a given time). So, the removal of single pixels in the range domain (before the integration along the rail) does not give necessarily blank pixels in the x–y image (even if sometimes it does).

IV. RADAR EQUIPMENT

A picture of the radar equipment is shown in Fig. 4. An IBIS-FM [13] by IDS Georadar has been used as the radar head. It operated a continuous-wave frequency modulation (CWFM) in the Ku-band.

The radar head was fixed on a rail carriage, moved by a step motor (model M60SH86 by AEC). The rail was also equipped by an incremental encoder (model Rotaplus I40 by Lika Elecronic) for detecting the position and the speed of the carriage. Table I resumes the main radar parameters.

The speed of the carriage is set in terms of pulse for second (pps). The acceleration and deceleration phases have to be set as fast as possible for maximizing the scan length. Indeed, only the data acquired at constant speed were considered for synthesizing the radar image.

The radar and the rail were not directly synchronized, and for this reason, the (apparent) displacement of a target close to the radar and the speed of rail were compared (see Fig. 5) to identify correctly the time interval at constant speed directly from the radar signal (also without relying on the encoder data).

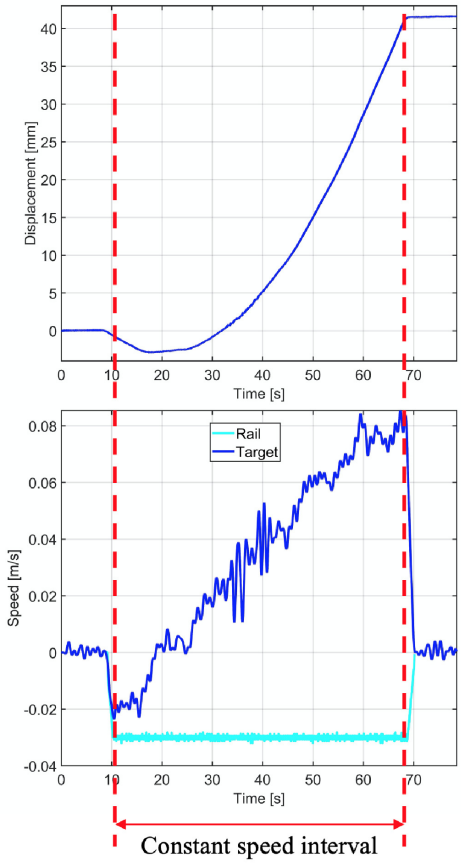


Fig. 5. Carriage speed detected by the encoder and by the radar signal of a target close to the radar.

The result of a single measurement is arranged in a matrix $E_{i,k}$ where the first index is the "fast time" (identifying each single frequency of the sweep) and the second index is the "slow time" (identifying each single-frequency sweep). A Kaiser window with $\beta = 7$ [14] is applied to the $E_{i,k}$ matrix along both the indexes. Finally, the FFT is calculated along the fast time (obtaining the range in the radar image) and along the slow time (obtaining the cross-range in the radar image).

To test the radar before the measurement campaign in front of a real quarry, a corner reflector was positioned at about 25 m in front of the radar. Fig. 6(a) shows the whole radar image including the high-frequency clutter. As the rail speed was 0.03 m/s, the corresponding Doppler frequency was $\Delta f_d = \pm 1.72$ Hz. Fig. 6(b) shows a zoom inside the Doppler band $\Delta f_d = \pm 1.72$ Hz. The corner reflector is clearly imaged in the right position. The target at about 36 m is a metallic fence about orthogonal to the radar field of view.

V. HIGH-FREQUENCY ACQUISITION METHOD: APPLICATION TO EXPERIMENTAL DATA

The radar technique proposed in this article has been tested in a realistic scenario: a quarry of inert materials in Tuscany,

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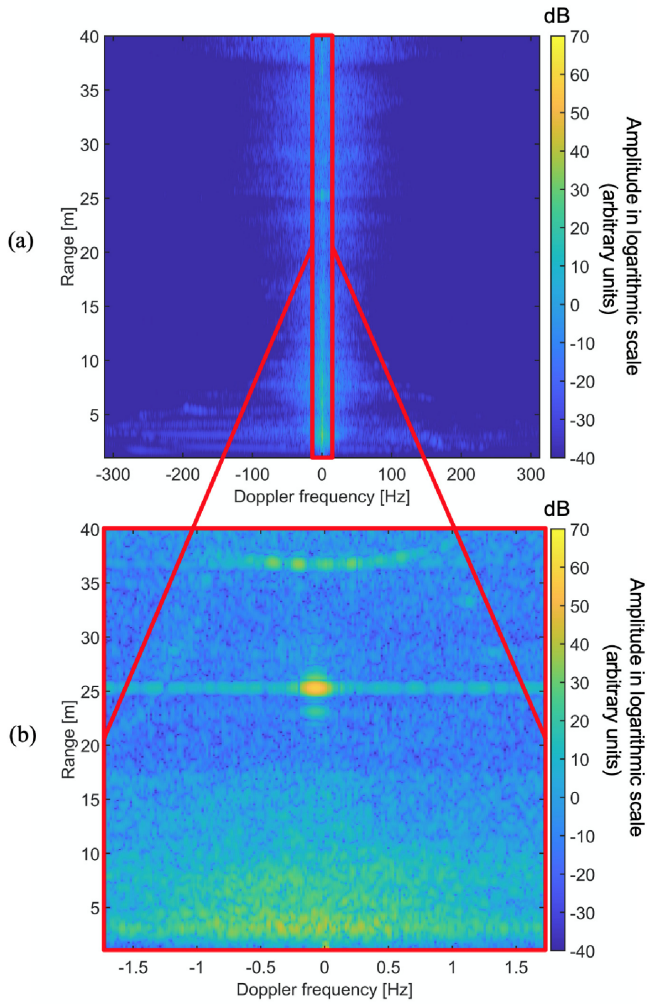


Fig. 6. Radar images of a single corner reflector 25 m in front of the radar. (a) Whole radar image including the high-frequency components. (b) Zoom inside the Doppler frequency given by the speed of the radar head on the rail.

TABLE II
RADAR PARAMETERS

Central frequency	17.20 GHz
Band	200 MHz
Field of view (-3dB)	50 °
Length of the rail	1.8 m
Sampling frequency	308 Hz
Radar-head speed along the rail	0.02 m/s

Italy. Fig. 7 shows the aerial view of the test site. The radar parameter set is reported in Table II.

The rail speed was 0.02 m/s (600 pps) corresponding to the Doppler frequency $\Delta f_d \simeq \pm 1.15$ Hz.

It is interesting to note that between the radar installation and the slope, there is a large metallic machinery, in particular, a crane 60-m long that was rotating during the radar acquisition.

Fig. 8 shows the image obtained by Fourier transform along the fast time (range) and along the slow time (cross-range).

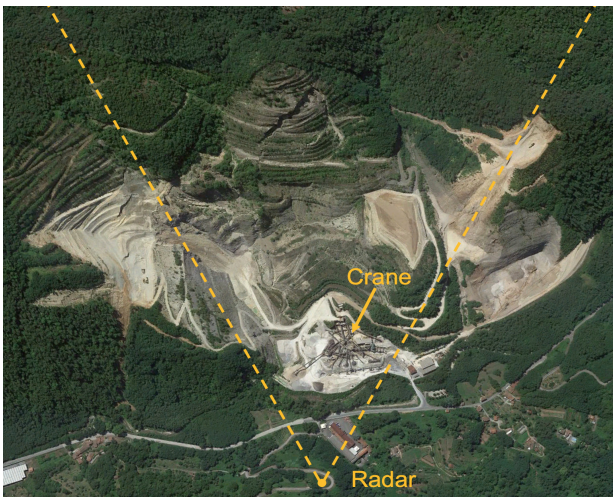


Fig. 7. Aerial image of the Pedogna test site showing the radar position, its field of view, and the metallic crane.

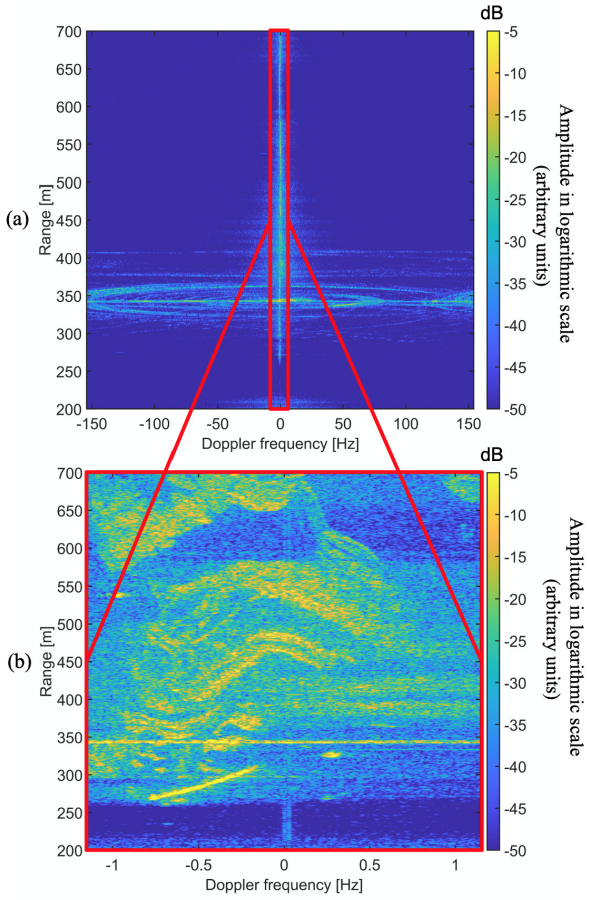


Fig. 8. Radar image of Pedogna quarry obtained sampling at 308 Hz the data acquired by a radar head moving at 0.02 m/s. The upper part (a) is the whole radar image including the high-frequency clutter and the lower part (b) is the zoom inside the Doppler frequency due to the speed of the radar head on the rail.

To obtain the image of the quarry is necessary to zoom inside the Doppler band of the rail $\Delta f_d \simeq \pm 1.15$ Hz.

In the radar image, two different zones can be identified: the first approximately between 320 and 370 m that is clearly affected by blurring due to the movement of a large rotating

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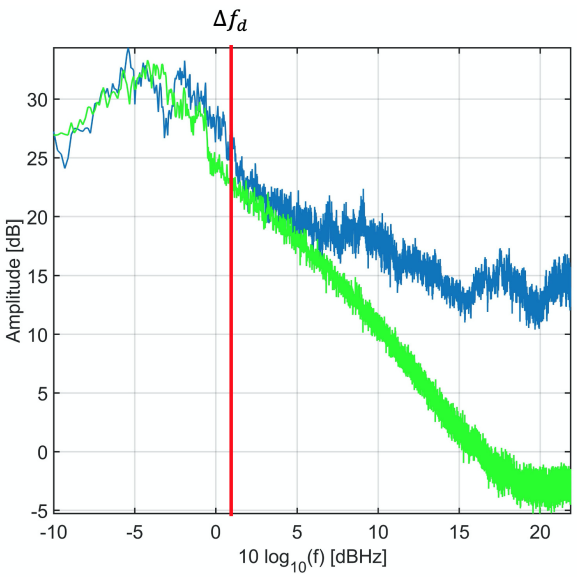


Fig. 9. Spectra in the range between 320 and 370 m (in blue) and in the range between 400 and 450 m (in green).

crane (its radius was about 60 m) and the rest of the image where the slope is excavated, but there are no large machineries in operation. The two zones give different spectra. Fig. 9 shows both the spectrum in the range between 320 and 370 m (in blue) and the spectrum in the range 400 and 450 m (in green). The two zones extend the same length in range (50 m) to be comparable in power.

In both the spectra, the maximum Doppler frequency (Δf_d) due to the movement of the rail is marked with a red line. The band $< \Delta f_d$ includes both the Doppler shift due to the rail and the possible clutter. The two contributions cannot be separated. The band $> \Delta f_d$ includes only clutter that can be cut off, just limiting the processed image between ± 1.15 Hz.

The radar image in Fig. 8(b) was obtained at the maximum sampling frequency allowed by the necessary unambiguous range for this scenario (with targets up to 2 km). To quantify how a lower sampling frequency worsens the quality of image, we resampled the data with the following decimation factors: 2, 4, 6, 8, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, and 70.

For each decimation factor, the quality of the image has been evaluated calculating the PSNR defined as [12]

$$\mathrm{PSNR} = \frac{\max \left| I_{i,j} \right|^2}{\frac{1}{mn} \sum_{i=1}^{m} \sum_{j=1}^{n} \left| I_{i,j} - R_{i,j} \right|^2} \tag{3}$$

where $I_{i,j}$ is the m $\times$ n image to be evaluated and $R_{i,j}$ is the reference image (the image acquired at the highest sampling rate). The inverse of PSNR can be considered a (rough) evaluation of the clutter energy (E_c) that re-folds inside the Doppler band as explained in Chapter I (Theory).

Fig. 10 shows the 1/PSNR versus sampling frequency for the image synthesized between 320 and 370 m (where there is the moving crane). The red line is $\propto 1/f_s$. It is noticeable that $E_c \approx 1/\mathrm{PSNR} \propto 1/f_s$ as predicted by the theory discussed above.

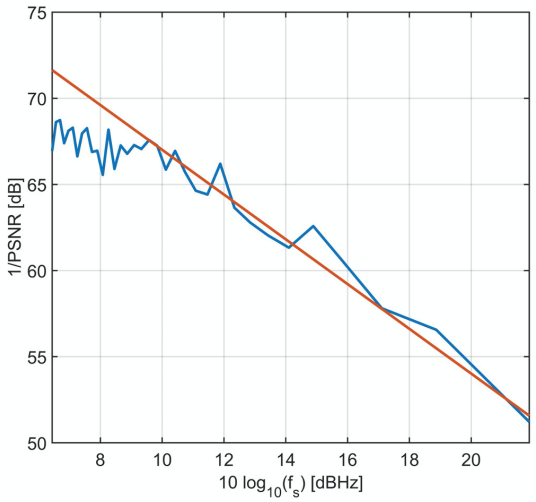


Fig. 10. 1/PSNR versus sampling frequency f_s in log scale for the zone between 320 and 370 m, where there is moving machinery.

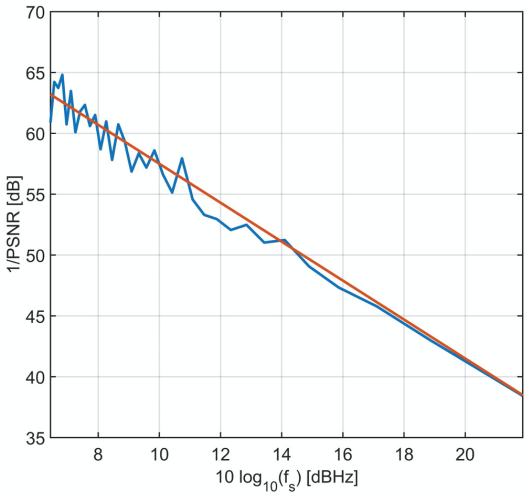


Fig. 11. 1/PSNR versus sampling frequency f_s in log scale for the zone between 400 and 700 m.

Fig. 11 shows the 1/PSNR versus sampling frequency for the image synthesized between 400 and 700 m. In this zone, the agreement with $1/f_s$ is even better.

The deterioration of the image quality at lower frequency can also be noted directly comparing the image acquired at 308 Hz [Fig. 8(b)] and the same image acquired at 4.4 Hz as shown in Fig. 12. This is the value of sampling frequency that allows to acquire along x with step smaller than $\lambda/4$. A sparser sampling produces angular aliasing in the image.

VI. Coherent Change Subtraction: Application to Experimental Data

High sampling frequency allows to cut off high-frequency clutter that could re-fold inside the Doppler band, but it can do anything for mitigating low-frequency clutter. For this reason, the authors even proposed a post-processing approach that is based on detection of changes between a couple of images taken at different times. It is a sort of "reverse" moving target indicator (MTI). The standard MTI rejects the static clutter,

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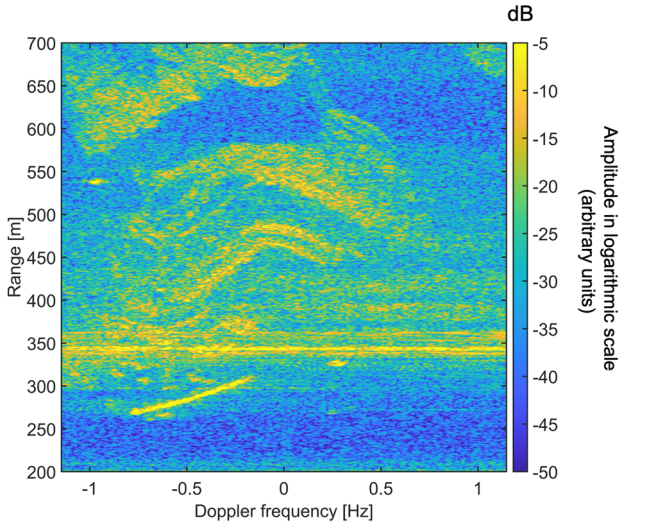


Fig. 12. Radar image of Pedogna quarry obtained sampling at 4.4 Hz the data acquired by a radar head moving at 0.02 m/s.

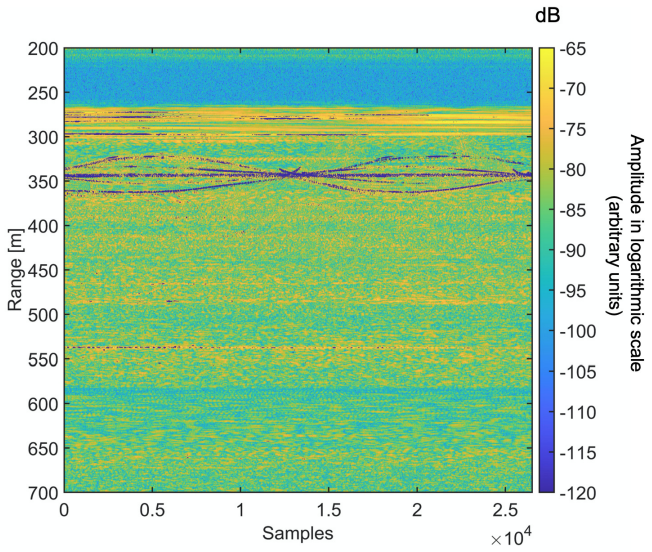


Fig. 13. Difference image between two images of Pedogna quarry processed only in short time (along the rows).

and this method rejects the moving clutter. Fig. 13 shows the image difference between two images processed only with an FFT along the short time (i.e., along the row of the matrix). The movements of the large crane at about 350 m gives characteristic artifacts that an amplitude in difference image is able to identify and cut off, putting the corresponding pixels to zero.

After this cleaning of the artifacts due to the movement of large and slow machineries, the image is also processed along the slow time (i.e., along the line of matrix in Fig. 15). The final result is shown in Fig. 14.

The image is much cleaner than the image acquired at low sampling frequency shown in Fig. 12; furthermore, the annoying blurring in the cross-range at about 350 m disappears. Therefore, this image is even better than the image in Fig. 8(b).

The procedure for removing the slow-moving clutter can also be applied successfully with data acquired by standard

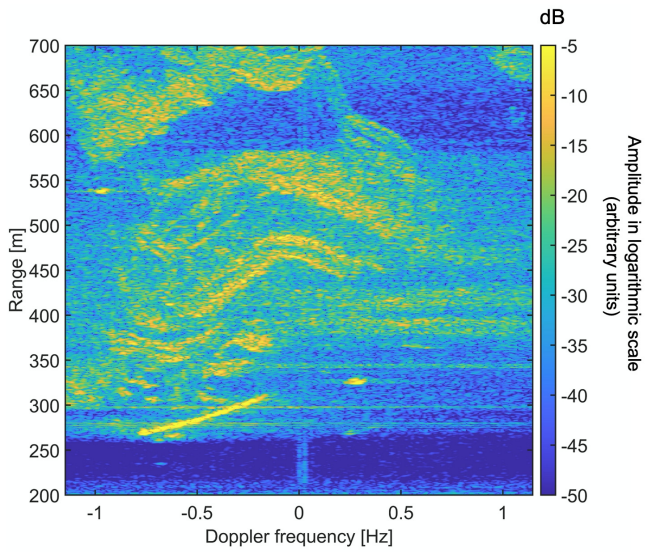


Fig. 14. Radar image of Pedogna quarry obtained sampling at 308 Hz the data acquired by a radar head moving at 0.02 m/s and after application of the procedure for removing the slow-moving clutter.

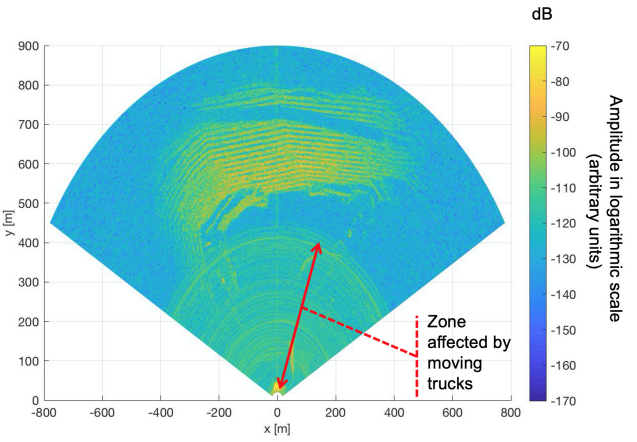


Fig. 15. Radar image of a large quarry in New Zealand affected by the movement of trucks up to 400 m range.

TABLE III
RADAR PARAMETERS

Central frequency	17.20 GHz
Band	200 MHz
Field of view (-3dB)	50 °
Length of the rail	1.8 m
Number of frequencies	2400
Number of positions along the rail	401

step-by-step GBSAR system and also with different focusing algorithms. To demonstrate it, the following case study is reported. In a large quarry in New Zealand, the continuous movement of the tracks affects all the ranges up to 400 m from the radar. Fig. 15 shows the radar image acquired with a step-by-step GBSAR. The radar parameters are reported in Table III.

The focusing algorithm was a back-propagation procedure as described in [7].

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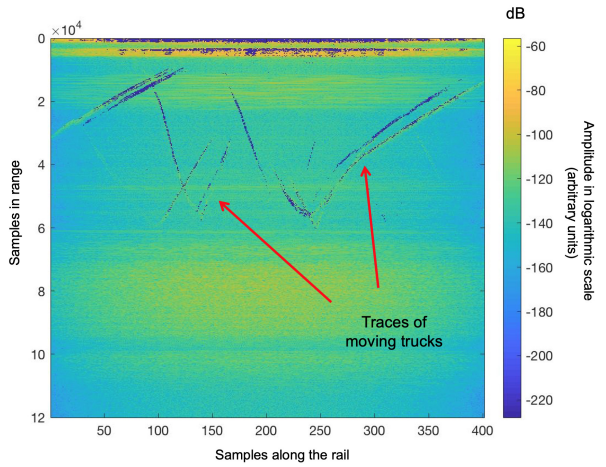


Fig. 16. Difference image between two images of a large quarry in New Zealand processed only in range, where the traces of the moving trucks are well-visible.

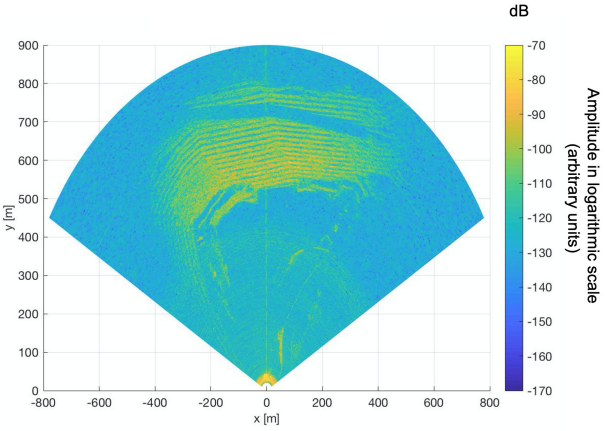


Fig. 17. Radar image of a large quarry in New Zealand processed after removal of the traces of the moving tucks.

Fig. 16 shows the image difference between two images processed only in range (i.e., along the row of the matrix). The traces of the tracks are clearly visible and can be removed with a threshold in the amplitude.

After removing the truck traces, the image is focused with a back-propagation algorithm. The result is shown in Fig. 17. The zone up to 400 now is much clearer.

VII. Conclusion

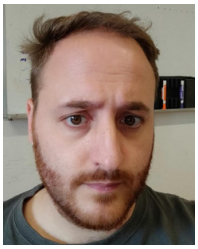
In this article, two different but complementary techniques have been proposed for mitigating the dynamic clutter in GBSAR images. The former is based on an acquisition modality named "free-running" different from the standard step-by-step acquisition, and the latter is a processing technique able to reject the low-frequency clutter that for its own nature cannot be filtered out by a fast acquisition. In this work, the clutter rejection technique has been implemented off-line, but as the processing involves only couples of subsequent images, it could be easily implemented on-line and so it could also be used in a real-time warning system.

The two techniques (high sampling frequency acquisition and coherent change subtraction) are complementary but independent. In particular, the later technique can also be successfully applied to the image acquired by standard step-by-step GBSAR.

Finally, the original aim that motivated this work was to mitigate the clutter in mining environment. Nevertheless, the same techniques could also be successfully applied in geotechnical application. For example, Frukacz and Wieser [15] reported the problem of the metallic fences in engineered landslides that prevent an effective monitoring by GBSAR. The authors believe that the free-running and clutter subtraction in post-processing can greatly improve the quality of the radar images in these and other similar cases.

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